Methods on which free cash flows are used

- • Net Present Value (NPV).
- • Discounted Payback Period.
- • Internal Rate Of Return (IRR).
- • Modified Internal Rate Of Return (MIRR).

NPV.

- NPV is the sum of the present values of all cash flows associated with a particular project.
- The discount rate used is the firms cost of capital.
- A project is viable if the NPV is positive.

Practice question on NPV.

- **A project requires an initial investment of K24,000 and will generate annual cash flows as follows:**

Year	**1**	**2**	**3**	**4**	**5**
Cash flows K	**7,800**	**6,000**	**4,200**	**7,400**	**9,200**

- **The cost of capital is 10%. Target payback period is 4 years**
- **<u>Required:</u>**
- **Determine the NPV and the discounted payback period.**

NPV solution :

Year	Cash flows K	Discount at 10%	Present values K
0	(24,000)	1.000	(24,000)
1	7,800	0.909	7,090
2	6,000	0.826	4,956
3	4,200	0.751	3,154
4	7,400	0.683	5,054
5	9,200	0.621	5,713
		NPV	1,967

Discounted payback period (DPP).

- **DPP is the exact length of time taken for the present values of cash inflows to be equated to the initial cost.**
- **A project is viable if its discounted payback period is less or equal to the company's target (if any) .**

Discounted payback period (DPP):

Year	Cash flows K	Discount at 10%	Present values K	Cumulative PVs
0	(24,000)	1.000	(24,000)	(24,000)
1	7,800	0.909	7,090	(16,910)
2	6,000	0.826	4,956	(11,954)
3	4,200	0.751	3,154	(8,800)
4	7,400	0.683	5,054	(3,746)
5	9,200	0.621	5,713	1,967
		DPP = 4 + 3746/5713 = 4.66 years		

Dealing with inflation.

- The impact of inflation can be dealt with in two different ways – both methods give the same NPV.
- The real method can only be used if all cash flows are inflating at the general rate of inflation.
- The money/ nominal method is more reliable in questions involving specific inflation rates for different variables.

Dealing with tax.

- **Main impacts of taxation in an investment appraisal are:**
- **1. Tax is charged on operating cash flows, and**
- **2. Tax allowable depreciation / capital allowances / writing down allowances can be claimed, thus generating tax relief**
- **In project appraisal the effects of taxation must be considered when calculating the free cash flows to be discounted.**

IRR of a project

- **It represents the discount rate at which the NPV of an investment is zero.**
- **Conventional projects (outflow followed by inflows) should be accepted if the IRR is greater than the firm's cost of capital.**
- ***The steps in linear interpolation are:***
- **(1) Calculate two NPVs for the project at two different costs of capital.**
- **(2) Use the following formula to find the IRR:**

Formula for Internal rate of return (IRR)

- $\text{IRR} = A + \left\{ \frac{a}{a - b} x \left(B - A \right) \right\}$
- Where;
- A = lower discount rate with positive NPV (a)
- B = Higher discount rate with negative NPV (b)

Formula for Internal rate of return (IRR)

- $IRR = A + \left\{ \frac{a}{a - b} x \left(B - A \right) \right\}$
- **Where;**
- **(A) r% = 5%, NPV = K32,000 (a)**
- **(B) r% = 10%, NPV = (K78,000) (b)**
- **Find IRR**
- **= 5% + {32,000/(32,000+78,000)*(10%-5%)} =6.45%**

Problems with IRR method of appraisal

- **1. The decision rule is not always clear cut.**
- **For example, if a project has 2 IRRs (or more), it is difficult to interpret the rule which says "accept the project if the IRR is higher than the cost of capital".**

Problems with IRR method of appraisal

- 2. IRR is often mistakenly assumed to be a measure of the return from a project, which it is not.
- The IRR only represents the return from the project if funds can be reinvested at the IRR for the duration of the project.

Problems with IRR method of appraisal

- **Since projects can have multiple IRRs (or none) it is difficult to meaningfully compare projects using IRR.**

- **Therefore, NPV of the project is considered more reliable for investment appraisal purposes.**

Modified Internal Rate of Return (MIRR).

- **The modified internal rate of return (MIRR) is the return on an investment, considering both the cash flows of the investment and the earnings on these cash flows based on a specific reinvestment rate.**

Steps in calculating Modified IRR.

- **<u>Step 1</u>: Calculate the present value of the outflows, discounted at the project's cost of capital. PV = FV/(1+r)^n**
- **<u>Step 2:</u> Calculate the future value of the inflows, compounded at the reinvestment rate . FV = P*(1 + r)^n**
- **<u>Step 3:</u> Calculate the return on the investment.**
- **<u>Step 4:</u> A project is acceptable if MIRR is greater than the projects cost of capital**

Modified IRR method of appraisal (MIRR)

- **MIRR is unique and simple percentage.**
- **Gives a measure of the return from a project**
- **Is popular with non-financially minded managers, as a simple rule can be applied:**
- **MIRR = Project's return**

Task

- **A project requires an initial investment of K10,000 and another additional investment of K2,000 at the end of year 2. It will generate annual cash flows as follows:**

Year	**1**	**3**	**4**	**5**
Cash flows K	**7,000**	**8,000**	**4,000**	**1,500**

- **The cost of capital is 10%. Funds can be reinvested at 11% per year**
- **<u>Required:</u>**
- **What is the MIRR?**

Calculating Modified IRR (MIRR)

- **The possible formula to use is:**

- $$\text{MIRR} = \left\{ \sqrt[n]{\frac{FVCF}{PVCF}} \right\} - 1$$

- **<u>where</u>**
- **FVCF = the future value of the "*return phase*" of the project**
- **PVCF = the present value of the "*investment phase*" of the project**

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Practice question for IRR and MIRR

- **A project requires an initial investment of K24,000 and will generate annual cash flows as follows:**

Year	**1**	**2**	**3**	**4**	**5**
Cash flows K	**7,800**	**6,000**	**4,200**	**7,400**	**9,200**

- **The cost of capital is 10%.**
- **<u>Required:</u>**
- **Determine the IRR of the project.**
- **Assume now that funds can be borrowed at 10% whilst the reinvestment rate for positive cash flows is 12%. What is the MIRR?**

IRR:

Year	Cash flow K	DF at 10%	PVs K	DF at 14%	PVs K
0	(24,000)	1.000	(24,000)	1.000	(24,000)
1	7,800	0.909	7,090	0.877	6,841
2	6,000	0.826	4,956	0.769	4,614
3	4,200	0.751	3,154	0.675	2,835
4	7,400	0.683	5,054	0.592	4,381
5	9,200	0.621	5,713	0.477	4,775
		NPV	1,967	NPV	(554)

IRR:

- A = 10%, a = 1,967, B = 14%, b = (554)
- IRR = 10% + [1,967/(1967+554)*(14%-10%)]
- = 10% + 3.12%
- = 13.12%

- The underlying assumption is that cash flows will be reinvested at IRR.
- Thus, at IRR, the project's NPV is zero

MIRR:

Year	Cash flow at year 0 K	Compound Factor $(1 + r)^n$	Equivalent cash flow at year 5 K
0	(24,000)		
1	7,800	1.12^4	12,273
2	6,000	1.12^3	8,430
3	4,200	1.12^2	5,268
4	7,400	1.12^1	8,288
5	9,200	1.12^0	9,200
		Total	43,459

MIRR:

- **Thus, what is the IRR of a cash outflow of K24,000 in year 0 followed by an inflow of K43,459 in year 5?**
- $\frac{43,459}{(1+r)^5} = 24,000$
- $\frac{24,000}{43,459} = (1 + r)^5$
- . MIRR = $\sqrt[5]{\left(\frac{43,459}{24,000}\right)} - 1 = 12.6\%$
- **This is therefore the return on the project.**
- **The project is acceptable since MIRR exceeds the cost of finance.**

Practice question 2 (Six-year project)

- **The initial outlay is K3 million and the expected cash flows originating from the project are K600, 000 for the first year, then increasing by K160, 000 each year in years 2, 3, 4, 5 and 6.**
- **The project will be financed purely by equity. The firm's equity beta is 1.25 while its asset beta (ungeared beta) is 0.90. The expected market return of LuSE securities is 6 percent while the average yield on government securities is 15 percent. The reinvestment rate is 8%.**
- **<u>REQUIRED:</u>**
- **What is the MIRR for the project?**

Practice question 3 (4-year project)

- **The initial outlay is K6 million followed by the following cash flows:**
- **Year 1 2 3 4**
- **Cash flows K 2,000,000 3,000,000 (1,500,000) 4,500,000**

- **the cost of capital is 11% while the reinvestment rate is 13%**

- **<u>REQUIRED:</u>**
- **What is the MIRR for the project?**